

Algorithms 2

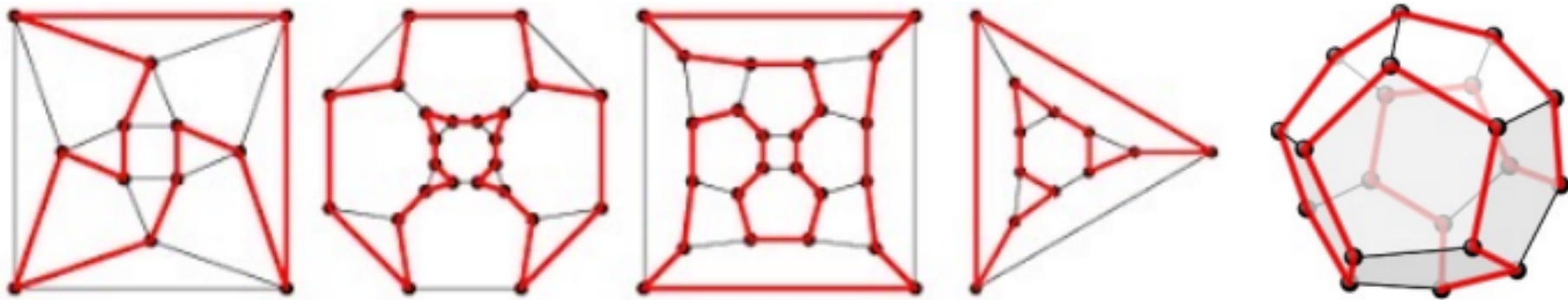
Hamiltonicity

Daniel Rosenberg & Michael Trushkin

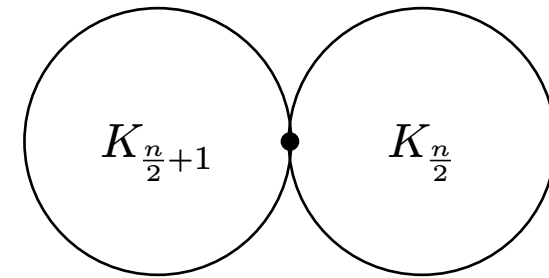
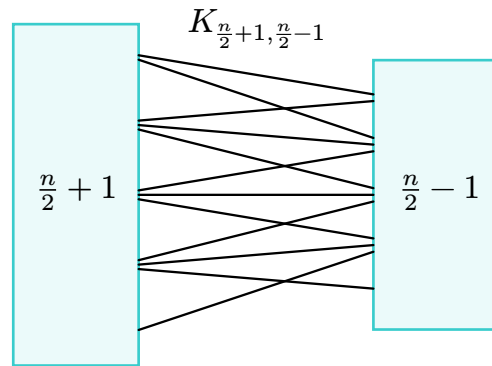
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1. Dirac's theorem

- A simple path spanning $V(G)$ is called a *Hamilton* path
- A simple cycle spanning $V(G)$ is called a *Hamilton* cycle
- A graph containing a Hamilton path is called *traceable*
- A graph containing a Hamilton cycle is called *Hamiltonian*



Here are two examples for **non**-hamiltonian graph



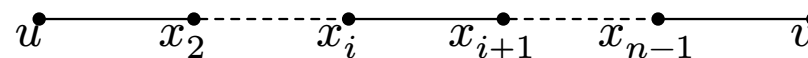
Question 1.1.1 What is common among the two examples?

Answer: They are both quite dense: $\delta(G) = \frac{n}{2} - 1$

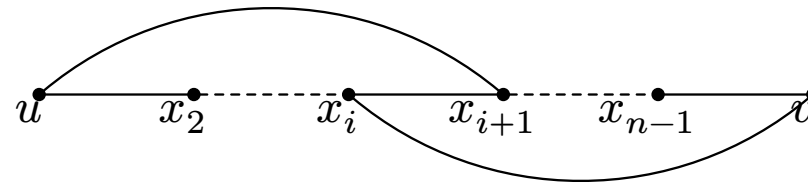
Theorem 1.1.2 (Dirac's theorem) Let G have $v(G) \geq 3$ and $\delta(G) \geq \frac{n}{2}$. then G is Hamiltonian

proof:

- Suppose the claim is false
- let G be an edge maximal counter example:
 - G does not contain an Hamiltonian cycle
 - $G + e$ is Hamiltonian $\forall e \notin E(G)$
- Let $e = uv \notin E(G)$
- As $G + e$ is Hamiltonian, G contains a Hamilton path

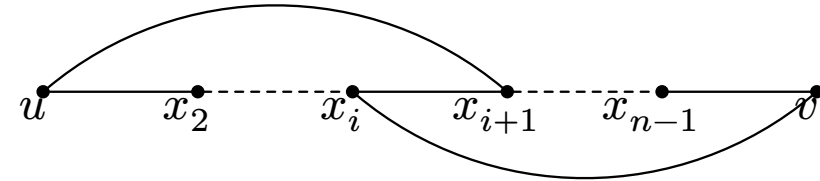


- If we could find an edge $x_i x_{i+1}$ such that
 - $ux_{i+1} \in E(G)$
 - $vx_i \in E(G)$
- Then we can reroute our path, and turn it into Hamilton cycle

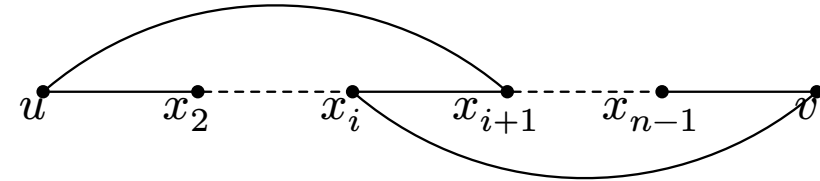


- Define $S := \{i \in [n] : ux_{i+1} \in E(G)\}$
 - note that $n \notin S$
 - Define $T := \{i \in [n] : vx_i \in E(G)\}$
 - note the $n \notin S$ as G is simple
- $\implies |S \cup T| \leq n - 1.$

-If $|S \cap T| \neq \emptyset$ we can reroute



- Define $S := \{i \in [n] : ux_{i+1} \in E(G)\}$
 - note that $n \notin S$
- Define $T := \{i \in [n] : vx_i \in E(G)\}$
 - note the $n \notin S$ as G is simple
- $\implies |S \cup T| \leq n - 1.$
- Assume the $|S \cap T| = \emptyset$

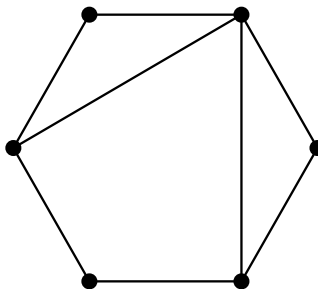


$$|S \cup T| = |S| + |T| - \underbrace{|S \cap T|}_0 \geq \deg(u) + \deg(v) \geq \frac{n}{2} + \frac{n}{2} = n$$

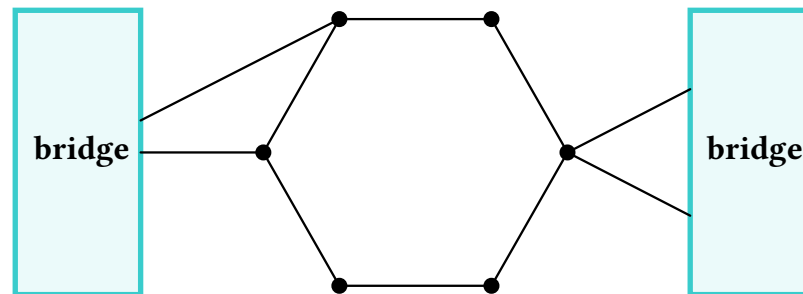
- We got $|S \cup T| \geq n$
- As $n \notin S \cup T$, we get a contradiction.

2. Eros-Chvatal

- Let C be a cycle in G .
- By a bridge of C we mean:
 1. an edge between two vertices of C , called a cord. **(those are trivial bridges)**

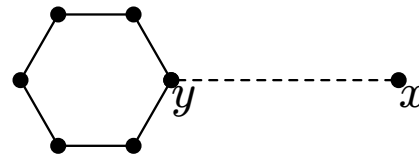


- connected components of $G - C$ that has neighbors in C , those are non-trivial bridges



Definition 2.1.1

- Let P be a $x \rightsquigarrow y$ path in G
- Let C be a cycle in G
- If $y \in C$ then $C \cup P$ is called (x, y) -lollipop

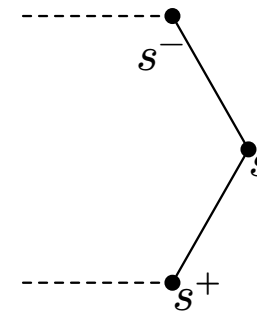


- Let $v \in V(C)$ and orient C
 - Denote by v^+ the successor of v
 - Denote by v^- the predecessor of v

For $S \subseteq V(C)$ write

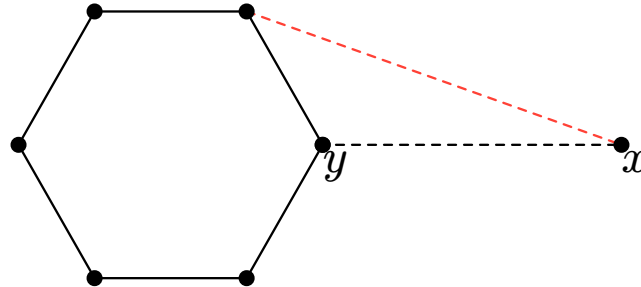
$$S^+ := \{s^+ : s \in S\}$$

$$S^- := \{s^- : s \in S\}$$



Lemma 2.1.2

- Let C be the largest cycle in G
- Let $C \cup P$ be an (x, y) -lollipop
- The path xy^+ and xy^- do not exist in G



Theorem 2.2.1 (Erdos Chvatal Theorem)

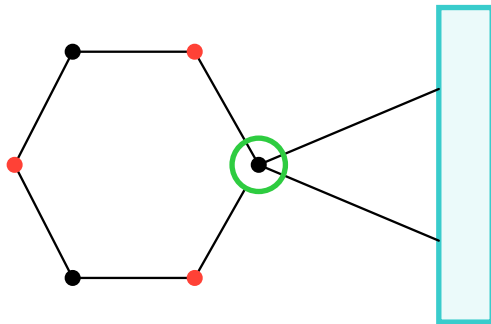
- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

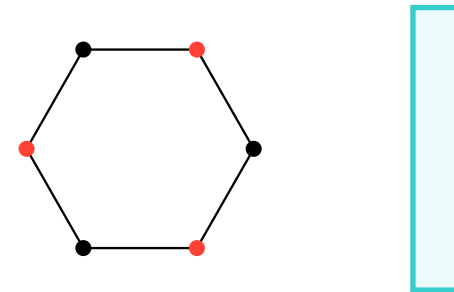
Theorem 2.2.1 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

proof:

- Assume the claim is false $\Rightarrow$ Let C be the largest cycle in G .
- Then G admits a non-trivial bridge with ≥ 2 points of attachment to C .
- Why?
 - Assume the bridge has < 2 points of attachments
 - Observe the components of $G - C$, one of the two can happen



$$1 = \kappa(G) \geq \alpha(G) \geq 2, \text{ Contridiction}$$


$$0 = \kappa(G) \geq \alpha(G) \geq 2, \text{ Contridiction}$$

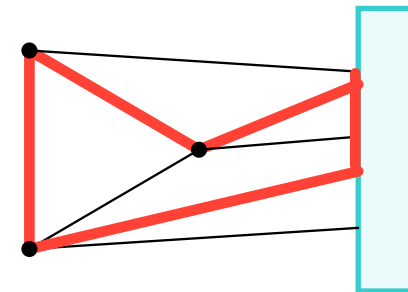
Assume the claim is false.

Goal : show that $\alpha(G) > \kappa(G)$
or show that a larger cycle exists.

Theorem 2.2.2 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

- C largest cycle in G .
- B the bridge, S the attachment points of B on C .
- Case 1: $V(C) = S$.
 - fix $x \in S$ and $b \in B$ then (x, b) -lollipop is such that $bx^+ \in E(G)$.
 - C can be extended, contradiction to the maximality of C .



Assume the claim is false.

Goal : show that $\alpha(G) > \kappa(G)$
or show that a larger cycle exists.

Theorem 2.2.3 (Erdos Chvatal Theorem)

- Let G have $v(G) \geq 3$
- If $\kappa(G) \geq \alpha(G)$ then G is Hamiltonian.

Case 2: $S \subset V(C)$:

- $|S| = \kappa(G)$.
- Every $b \in B$ and $s \in S$ form a (b, s) -lollipop
- Orient $C \Rightarrow S^+$ is independent
 - Otherwise C can be extended
- By the lollipop lemma: $\forall b \in B, s \in S : bs^+ \notin E(G)$
 - $\Rightarrow$ The set $S^+ \cup \{b\}$ is independent for every $b \in B$.
- $\alpha(G) \geq |S| + 1 = \kappa(G) + 1$.

